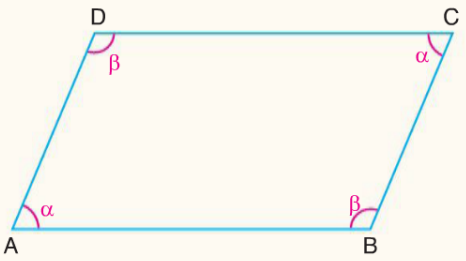
Paralelkenar

Karşılıklı kenarları paralel olan dörtgene paralelkenar denir. Paralelkenarda karşılıklı açıların ölçüleri eşit, komşu açılar bütünlerdir.



$m(\widehat{DAB}) = m(\widehat{BCD}) = \alpha$

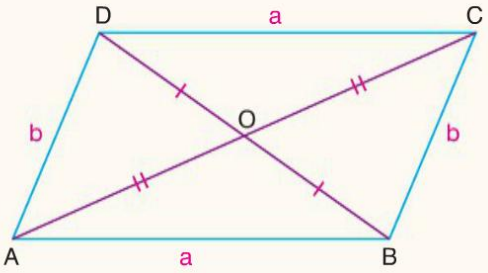
$m(\widehat{ADC}) = m(\widehat{ABC}) = \beta$

$\alpha + \beta = 180°$

★ Paralelkenarda karşılıklı kenarların uzunlukları birbirine eşittir.

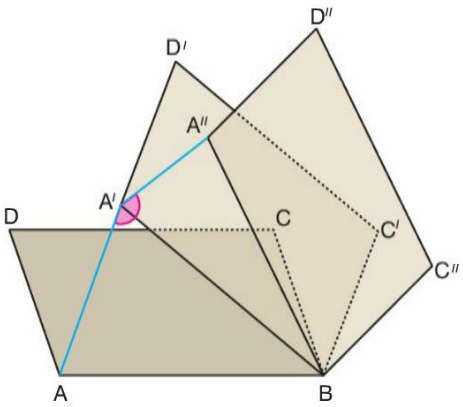
★ Paralelkenarda köşegen uzunlukları birbirine eşit değildir.

★ Paralelkenarda köşegenler birbirini ortalar.



Çevre(ABCD) = 2(a+b)

ÖR$_1$:

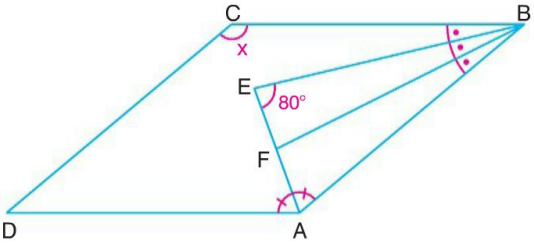


ABCD paralelkenarı biçimli bir karton negatif yönde B köşesi etrafında önce 40° döndürülerek daha sonra 24° döndürülerek $BA'D'C'$ ve $BA''D''C''$ paralelkenarı elde ediliyor.

Buna göre, $m(\widehat{AA'A''})$ açısı kaç derecedir?

A) 100 B) 110 C) 120 D) 136 E) 148

ÖR$_2$:



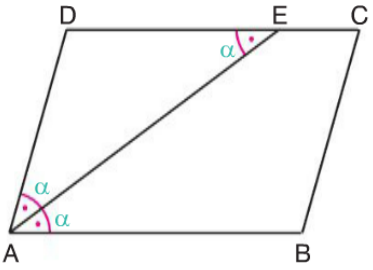
ABCD paralelkenar, $m(\widehat{AEB}) = 80°$

Şekilde verilen [AE] doğru parçası BAD açısını iki eşit açıya, [BE] ve [BF] doğru parçaları ise ABC açısını üç eşit açıya bölmektedir.

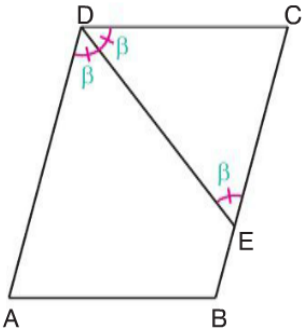
Yukarıdaki verilere göre, $m(\widehat{DCB}) = x$ kaç derecedir?

A) 115° B) 120° C) 130° D) 135° E) 140°

✦ Bir paralelkenarın herhangi bir köşesinden açıortay çizilmişse ikizkenar üçgen vardır.

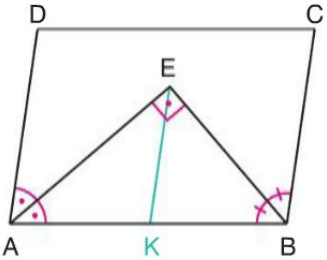


$|AD| = |DE|$



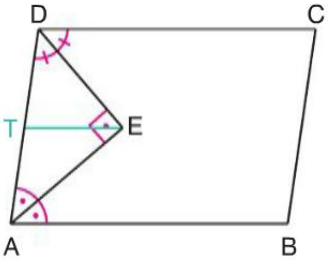
$|CD| = |EC|$

✦ Bir paralelkenarın komşu iki açısından açıortay çizilmişse muhteşem üçlü vardır.



$[AE] \perp [BE]$

$|AK| = |BK| = |EK|$



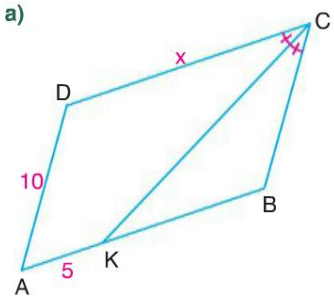
$[AE] \perp [DE]$

$|DT| = |AT| = |TE|$

ÖR$_3$:

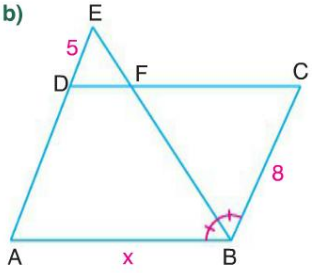
Aşağıda verilen ABCD paralelkenarında x değerini bulunuz.

a)



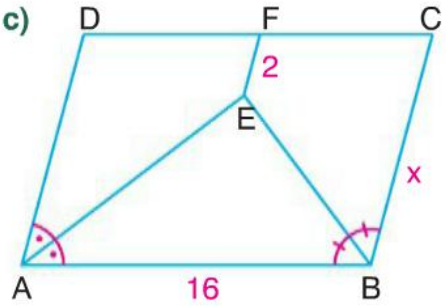
x=.....

b)



x=.....

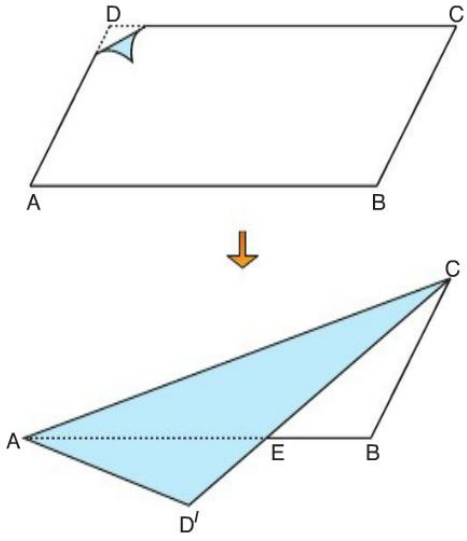
c)



x=.....

ÖR$_4$:

Çevresi 54 birim olan ABCD paralelkenarı biçimindeki kağıt [AC] köşegeni boyunca katlandığında D köşesi D′ noktasına gelmektedir.



[AB]∩[CD′]={E} olduğuna göre, AD′E üçgeninin çevresi kaç birimdir?

A) 18 B) 24 C) 27 D) 30 E) 36

Not:

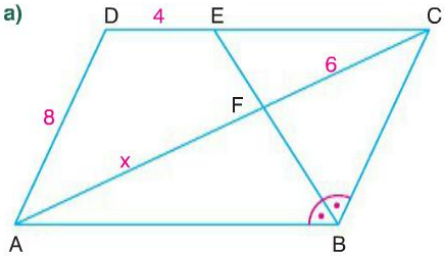
Paralelkenarın karşılıklı kenarlarının paralel olduğunu biliyoruz. Bu paralellik bize Thales ve Kelebek Benzerliği kullanma imkanı sağlar.

Uzunluk sorularında Pisagor kullanamıyorsanız, Thales ve Kelebek Benzerliği aklınızda olsun!

ÖR$_5$:

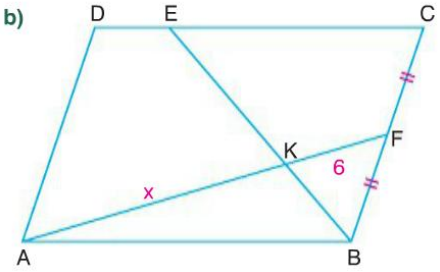
Aşağıda verilen ABCD paralelkenarında x değerini bulunuz.

a)



x=.....

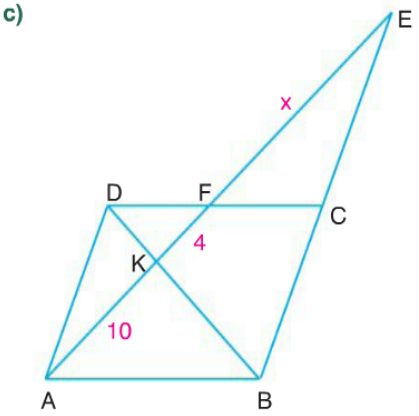
b)



|EC|=3|ED|

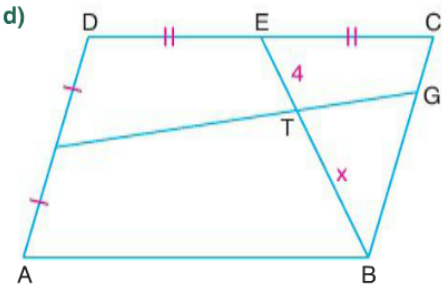
x=.....

c)



x=.....

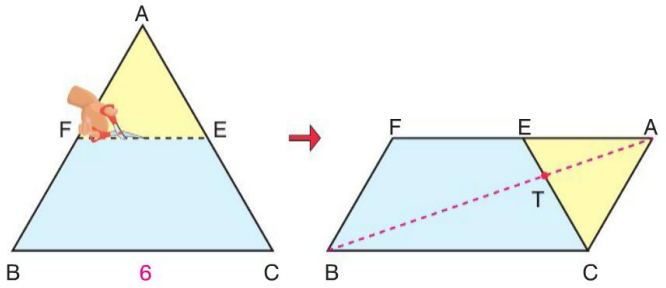
d)



3|CG|=|BG|

x=.....

ÖR$_6$:

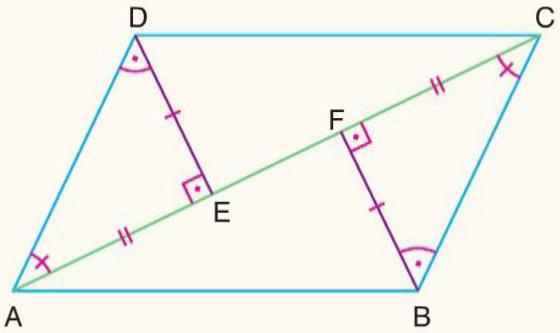


Bir kenarı 6 birim olan eşkenar üçgen biçimindeki karton E ve F orta noktalarından kesilerek oluşan AEF üçgeni yandaki gibi [EC] kenarına yapıştırılarak AFBC paralelkenarı oluşturuluyor.

[AB]∩[EC]={T} olduğuna göre, |AT| kaç birimdir?

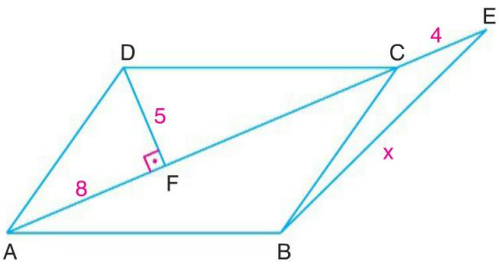
A) $\sqrt{5}$ B) $\sqrt{6}$ C) $\sqrt{7}$ D) $2\sqrt{2}$ E) 3

ABCD paralelkenarında [AC] köşegen, [DE]⊥[AC] ve [BF]⊥[AC] ise



|DE|=|BF| ve |AE|=|CF| dir.

ÖR$_7$:



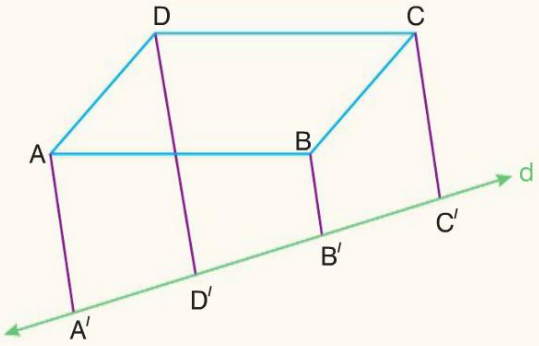
ABCD paralelkenar, EAB üçgen, [DF]⊥[AE], |AF|=8 cm

|DF|=5 cm, |CE|=4 cm

Yukarıdaki verilere göre, |BE|=x kaç cm dir?

A) $4\sqrt{3}$ B) 13 C) $6\sqrt{5}$ D) $10\sqrt{2}$ E) 15

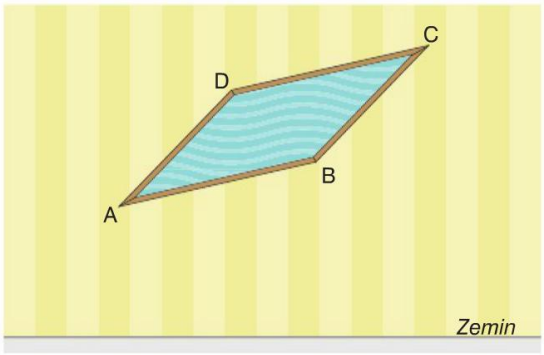
d doğrusu ABCD paralelkenarının dışında herhangi bir doğru, [AA′]//[DD′]//[BB′]//[CC′] olmak üzere;



|AA′|+|CC′|=|DD′|+|BB′| eşitliği vardır.

ÖR$_8$:

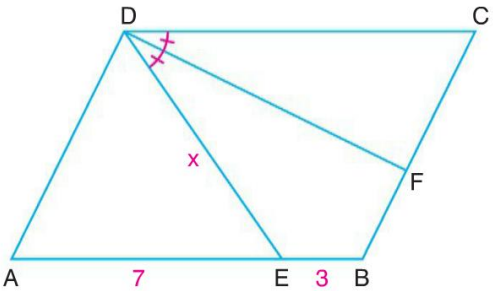
Aşağıdaki şekilde A, B ve C köşelerinin zemine uzaklığı sırasıyla 64, 68 ve 76 cm olan ABCD paralelkenar biçimindeki tablo gösterilmiştir.



Yukarıdaki verilere göre, tablonun D köşesinin zemine uzaklığı kaç cm'dir?

A) 64 B) 66 C) 68 D) 70 E) 72

ÖR$_9$:



ABCD paralelkenar, [DF] açıortay, |CF|=2|FB|

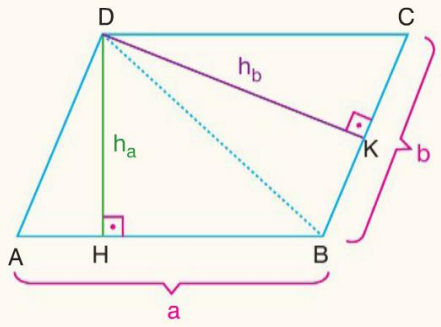
|AE|=7 cm, |EB|=3 cm

Yukarıdaki verilere göre, |DE|=x kaç cm dir?

A) 6 B) 7 C) 8 D) 9 E) 10

Paralelkenar da alan

Paralelkenarın alanı, bir kenar uzunluğu ile bu kenara ait yüksekliğin çarpımına eşittir.



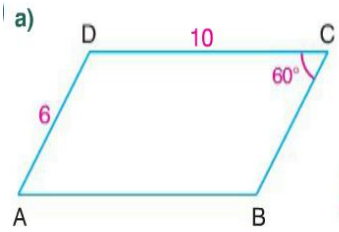
$Alan(ABCD) = a.h_a$

$= b.h_b$

$ÖR_{10}$:

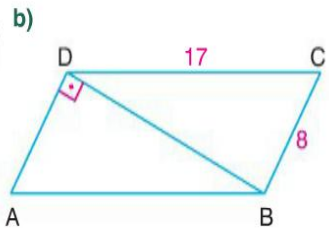
Aşağıda verilen paralelkenarların alanını hesaplayınız.

a)



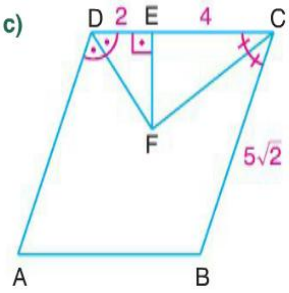
Alan(ABCD)=.....

b)



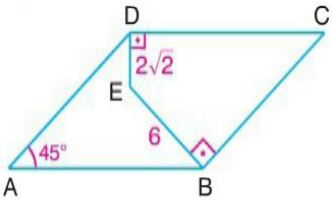
Alan(ABCD)=.....

c)



Alan(ABCD)=.....

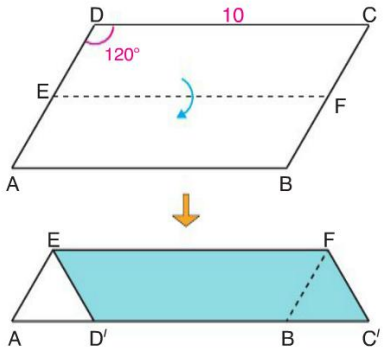
d)



Alan(ABCD)=.....

$ÖR_{11}$:

Bir kenarı 10 birim olan ABCD paralelkenarı biçimindeki kâğıt parçası EF boyunca katlandığında AB ile CD kenarları çakışmaktadır. Katlama işlemi sonunda oluşan AC′FE yamuğunun çevresi 29 birimdir.

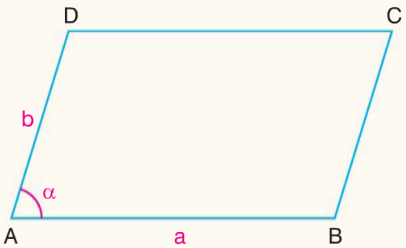


$m(\widehat{ADC})=120°$ olduğuna göre, başlangıçta verilen ABCD paralelkenarının alanı kaç birimkaredir?

A) $30\sqrt{3}$ B) $24\sqrt{5}$ C) $32\sqrt{3}$ D) $18\sqrt{10}$ E) $28\sqrt{5}$

İki Kenarı ve Bir Açısı Verilen Paralelkenarın Alanı

$|AB|=a,\ |AD|=b,\ m(\widehat{DAB})=\alpha$

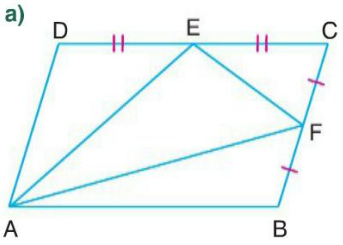


$Alan(ABCD) = a.b.\sin\alpha$

$ÖR_{12}$:

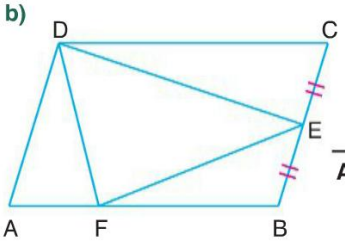
Aşağıda verilen sorularda istenilen oranları hesaplayınız.

a)



$$\frac{Alan(AEF)}{Alan(ECF)} = ...$$

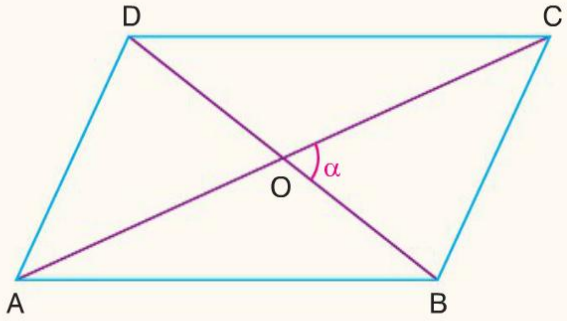
b)



$2|AF| = |BF|$

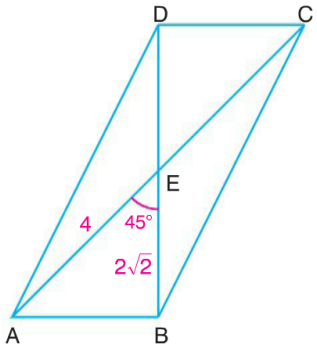
$$\frac{Alan(DEF)}{Alan(ECD)+Alan(EFB)} = ...$$

$|AC|=e,\ |BD|=f,\ m(\widehat{COB})=\alpha$



$$\text{Alan}(ABCD)=\frac{1}{2}\cdot e\cdot f\cdot \sin\alpha$$

ÖR$_{13}$:



ABCD paralelkenar

[AC] ve [BD] köşegen

$m(\widehat{AEB})=45°$

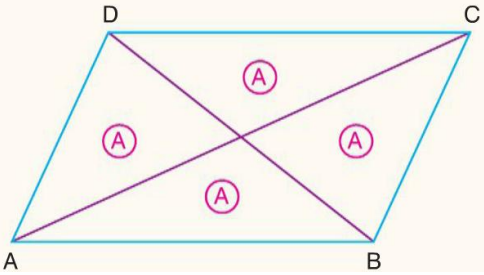
$|AE|=4$ cm

$|BE|=2\sqrt{2}$ cm

Yukarıdaki verilere göre, Alan(ABCD) kaç cm^2 dir?

A) 12 B) 16 C) 18 D) 20 E) 24

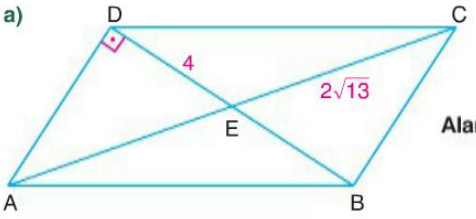
Bir paralelkenarda köşegenler paralelkenarı dört eşit alanlı üçgene ayırır.



ÖR$_{14}$:

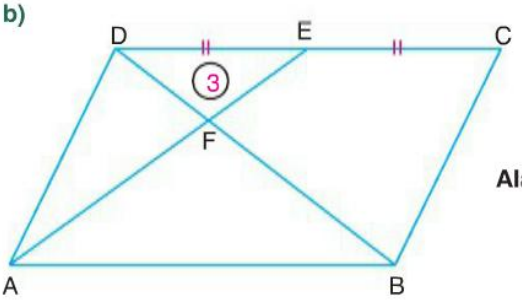
Aşağıda verilen paralelkenarlarda istenilen bölgelerin alanlarını hesaplayınız.

a)



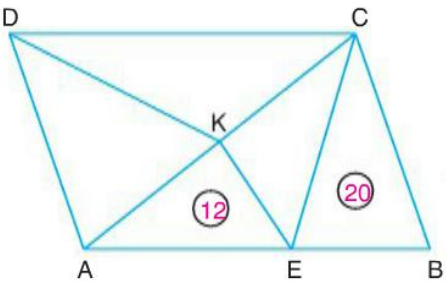
Alan(ABCD)=.....

b)



Alan(EFBC)=...

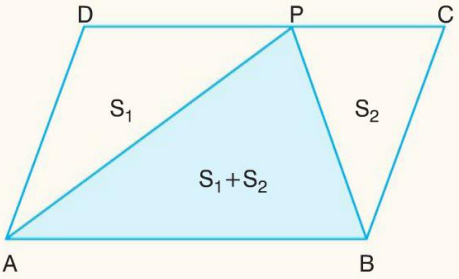
c)



$2|AE|=3|EB|$

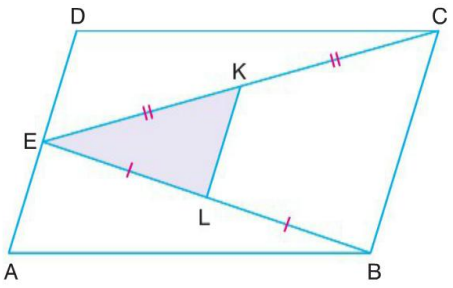
Alan(AKD)=.....

ABCD paralelkenarında $P\in[DC]$, S_1 ve S_2 bulundukları bölgelerin alanları



$$\text{Alan}(PAB)=\frac{\text{Alan}(ABCD)}{2}=S_1+S_2$$

ÖR$_{15}$:

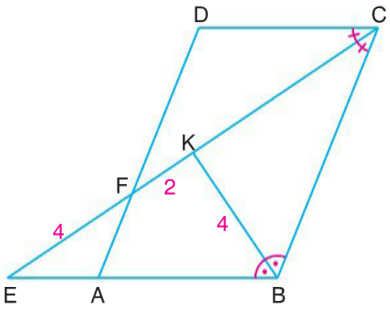


ABCD paralelkenar, EBC üçgen, $|BL|=|LE|$
$|CK|=|KE|$, Alan(ABCD)$=48$ cm^2

Yukarıdaki verilere göre, Alan(EKL) kaç cm^2 dir?

A) 6 B) 7 C) 8 D) 9 E) 10

ÖR$_{16}$:



ABCD paralelkenar
CEB üçgen
[BK] ve [CE] açıortay
$|EF|=|BK|=4$ cm
$|FK|=2$ cm

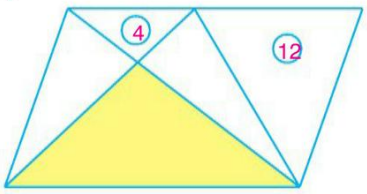
Yukarıdaki verilere göre, Alan(ABCD) kaç cm^2 dir?

A) 24 B) 28 C) 30 D) 32 E) 36

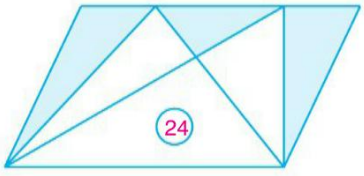
ÖR$_{17}$:

Aşağıda verilen paralelkenarlarda boyalı bölgelerin alanını bulunuz.

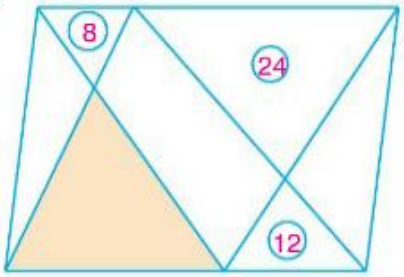
a)



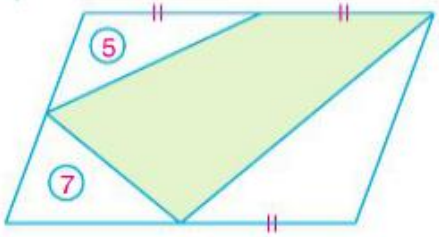
b)



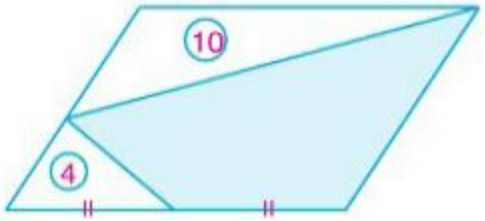
c)



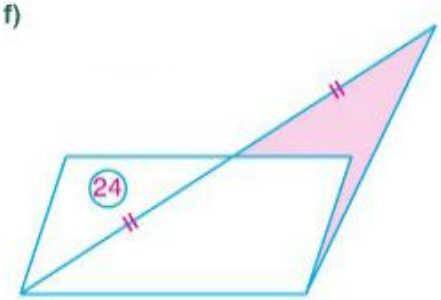
d)



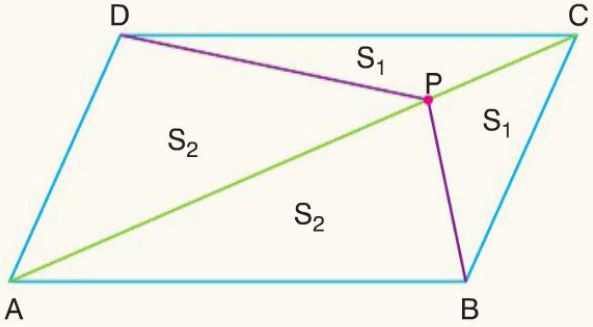
e)



f)

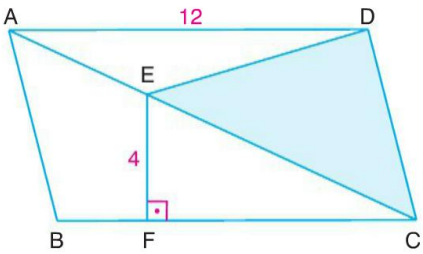


ABCD paralelkenarında [AC] köşegen ve P∈[AC]



Alan(DPC)$=$Alan(CPB)$=S_1$
Alan(DAP)$=$Alan(PAB)$=S_2$

ÖR$_{18}$:

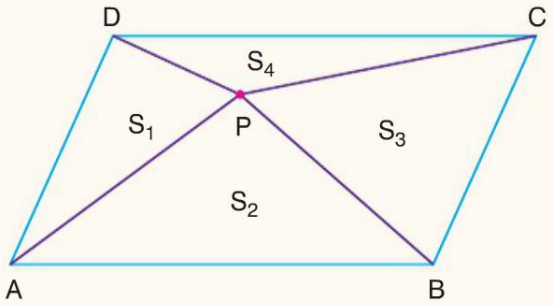


ABCD paralelkenar
[AC] köşegen
[EF] ⊥ [BC]
|AD| = 12 birim
|EF| = 4 birim

Yukarıdaki verilere göre, Alan(DEC) kaç birimkaredir?

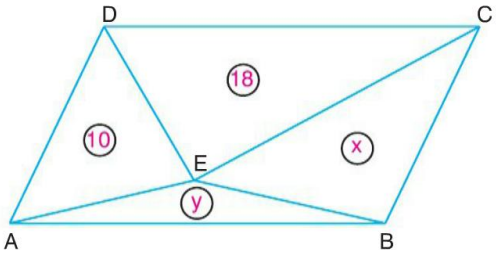
A) 20 B) 24 C) 30 D) 32 E) 36

P, ABCD paralelkenarının iç bölgesinde herhangi bir nokta ve S_1, S_2, S_3, S_4 bulundukları bölgelerin alanları



$$S_1 + S_3 = S_2 + S_4 = \frac{\text{Alan}(ABCD)}{2}$$

ÖR$_{19}$:

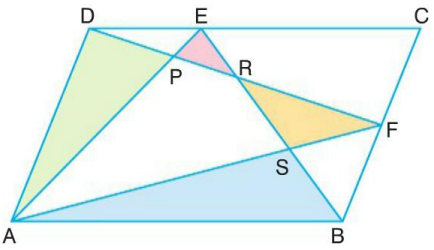


ABCD paralelkenar, Alan(ADE) = 10 cm^2, Alan(DEC) = 18 cm^2
Alan(EBC) = x cm^2, Alan(EAB) = y cm^2

Yukarıdaki verilere göre, x – y kaçtır?

A) 4 B) 5 C) 6 D) 8 E) 10

ÖR$_{20}$:

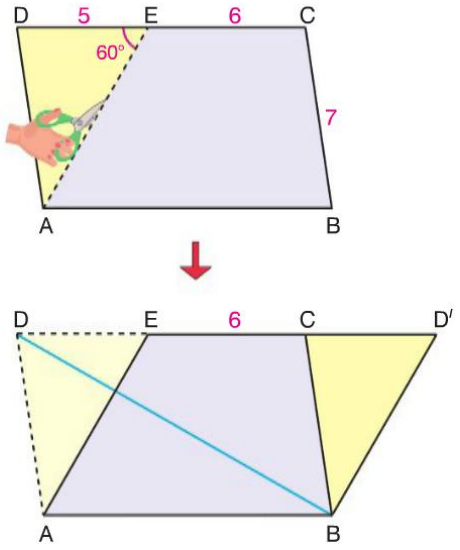


ABCD paralelkenar, ADF ve AEB üçgen
Alan(PER) + Alan(ASB) = 17 cm^2, Alan(DPA) = 12 cm^2

Yukarıdaki verilere göre, Alan(RSF) kaç cm^2 dir?

A) 2 B) 3 C) 4 D) 5 E) 6

ÖR$_{21}$:

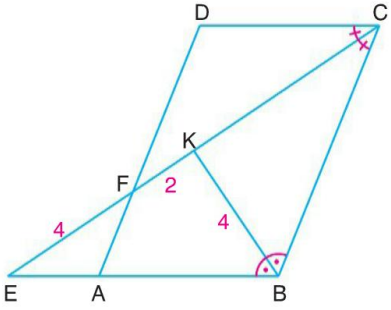


ABCD paralelkenarı şeklindeki kâğıt, m($\widehat{DEA}$) = 60° olacak şekilde [AE] boyunca kesilip [CB] ile [AD] kenarları çakışacak biçimde yukarıda gösterildiği gibi yapıştırılıyor.

Buna göre, Alan(DBD′) kaç birimkaredir?

A) $24\sqrt{3}$ B) 48 C) $32\sqrt{3}$ D) $36\sqrt{3}$ E) 60

ÖR$_{22}$:

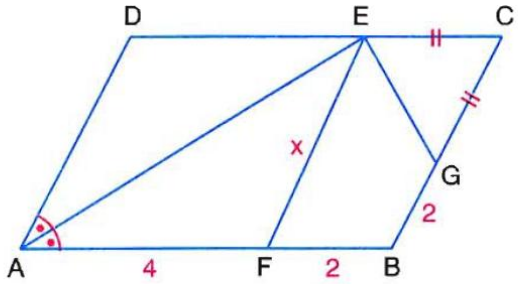


ABCD paralelkenar
CEB üçgen
[BK] ve [CE] açıortay
|EF| = |BK| = 4 cm
|FK| = 2 cm

Yukarıdaki verilere göre, Alan(ABCD) kaç cm^2 dir?

A) 24 B) 28 C) 30 D) 32 E) 36

ÖR$_{23}$:



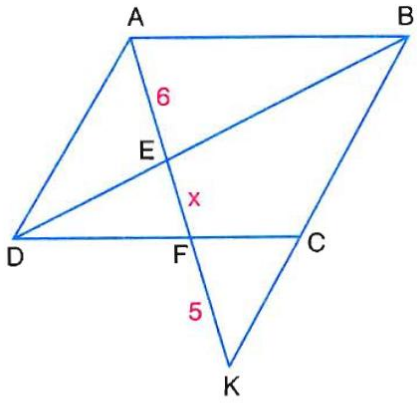
ABCD paralelkenar, $m(\widehat{DAE}) = m(\widehat{EAB})$, $|EC| = |CG|$

$|BG| = |BF| = 2$ cm, $|AF| = 4$ cm

Yukarıdaki verilere göre, $|EF| = x$ kaç cm dir?

A) 7 B) 6 C) 5 D) 4 E) 3

ÖR$_{24}$:



ABCD paralelkenar

$[BD] \cap [AK] = \{E\}$

B, C, K doğrusal

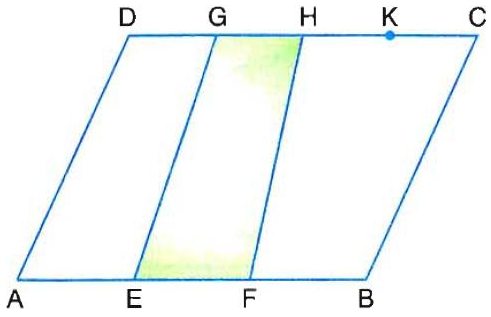
$|AE| = 6$ cm

$|FK| = 5$ cm

Yukarıdaki verilere göre, $|EF| = x$ kaç cm dir?

A) 7 B) 6 C) 5 D) 4 E) 3

ÖR$_{25}$:

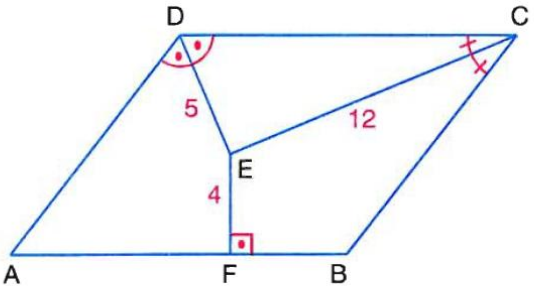


Şekildeki ABCD paralelkenarının [AB] kenarı üç eşit parçaya, [DC] kenarı ise dört eşit parçaya bölünmüştür.

Alan(EFHG) = 14 cm^2 olduğuna göre, Alan(ABCD) kaç cm^2 dir?

A) 42 B) 48 C) 50 D) 54 E) 60

ÖR$_{26}$:



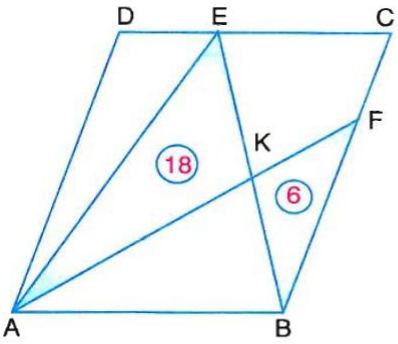
ABCD paralelkenar, $[EF] \perp [AB]$, [DE] ve [CE] açıortay

$|DE| = 5$ cm, $|CE| = 12$ cm, $|EF| = 4$ cm

Yukarıdaki verilere göre, Alan(ABCD) kaç cm^2 dir?

A) 112 B) 108 C) 96 D) 90 E) 84

ÖR$_{27}$:



ABCD paralelkenar

$[AF] \cap [BE] = \{K\}$

$|BF| = 2|FC|$

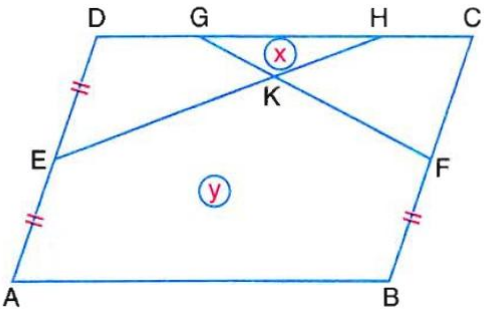
Alan(AEK) = 18 cm^2

Alan(BKF) = 6 cm^2

Yukarıdaki verilere göre, Alan(ABCD) kaç cm^2 dir?

A) 48 B) 54 C) 60 D) 64 E) 72

ÖR$_{28}$:



ABCD paralelkenar, $[EH] \cap [GF] = \{K\}$

$|AE| = |ED| = |BF|$, $|AB| = 2|GH|$

x ve y taralı bölgelerin alanları olduğuna göre, $\frac{x}{y}$ oranı kaçtır?

A) $\frac{1}{16}$ B) $\frac{1}{15}$ C) $\frac{1}{12}$ D) $\frac{1}{10}$ E) $\frac{1}{8}$